

TEST CASE REPORT

TC-01: Mentor Matching & Booking

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1. Test Case Identification

Test Case ID	TC-01
Test Case Title	Mentor Matching & Booking
Requirement ID	FR-001, FR-002, FR-003, FR-004
Requirement Type	Functional Requirement
Test Category	Core Functionality / Matching & Booking
Priority	High
Test Owner	Nishkal M R

2. Requirement Under Test

FR-001: The system shall recommend alumni mentors to students based on domain matching, such as Embedded Systems, Cloud Computing, AI/ML, or other selected areas.

FR-002: The system shall allow alumni mentors to specify their available dates and time slots and allow students to view those available slots before booking a session.

FR-003: The system shall allow students to book a mentorship or mock interview session with a selected alumni mentor using an available time slot.

FR-004: The system shall add a confirmed mentorship session to the calendars of both the student and alumni mentor and provide a meeting link for the session.

3. Test Objective

To verify that the system recommends alumni mentors to a student based on accurate domain matching, displays a mentor's currently available time slots, and successfully creates a confirmed booking that is added to both users' calendars with a valid meeting link.

4. Test Scope

In Scope:

- Domain-based recommendation of alumni mentors to a student.
- Exclusion of mentors whose domain does not match the student's selected domain.
- Display of a mentor's current, unoccupied available time slots to the student.
- Booking of a mentorship or mock interview session in an available slot.
- Addition of a confirmed session to both the student's and mentor's calendars.
- Generation and inclusion of a valid meeting link for the booked session.

Out of Scope:

- Prevention of double-booking of a mentor's slot (covered under TC-02).
- Recording of post-interview scorecards (covered under TC-03).
- Privacy protection and soft-deletion of user data (covered under TC-04).
- System response time and behaviour under simulated peak load (NFR-002).

5. Preconditions

1. The system is available for testing.
2. A valid student account exists with a selected domain of interest (e.g., Cloud Computing).

3. At least one alumni mentor is registered with domain expertise matching the student's selected domain and has published available time slots.
4. At least one alumni mentor is registered with a non-matching domain, for use as a negative check.
5. The mentor has marked specific slots as available and other slots as unavailable/occupied.
6. The calendar synchronization and meeting-link generation services are functional.

6. Test Data

Test Data	Description	Purpose
Student Account	A student account with domain preference set to Cloud Computing.	Used to request mentor recommendations and book a session.
Matching Mentor Profile	Alumni mentor profile with domain expertise in Cloud Computing and published available slots.	Used to verify correct domain matching and slot display.
Non-Matching Mentor Profile	Alumni mentor profile with domain expertise in Embedded Systems.	Used to verify that non-matching mentors are excluded from recommendations.
Available Time Slot	A specific time slot published as available by the matching mentor.	Used to verify accurate slot display and to complete a booking.
Booking Request	A valid request to book a mentorship or mock interview session in an available slot.	Used to trigger session creation, calendar sync, and meeting-link generation.

7. Test Procedure

Step	Action	Expected Result
1	Log in as the student.	Authentication succeeds.
2	Select a domain (e.g., Cloud Computing) and request mentor recommendations.	The system displays alumni mentors whose expertise matches the selected domain.
3	Review the recommended mentor list.	The non-matching mentor (e.g., Embedded Systems) does not appear in the list.
4	Select the matching mentor and view their available time slots.	Only the mentor's current, unoccupied available slots are displayed.
5	Select an available slot and initiate a booking.	A booking confirmation screen is presented for the selected slot.
6	Confirm the booking.	The system creates a confirmed mentorship session for the selected slot.
7	Check the student's calendar.	The confirmed session appears on the student's calendar with a valid meeting link.

8	Check the mentor's calendar.	The confirmed session appears on the mentor's calendar with a valid meeting link.
9	Open the meeting link from either calendar entry.	The link resolves to a valid, functional meeting room.
10	Re-check the mentor's available slots as a new student would view them.	The booked slot no longer appears as available.

8. Expected Results

- The system recommends only alumni mentors whose domain matches the student's selected domain.
- Mentors with a non-matching domain are excluded from the recommendation list.
- The mentor's currently available time slots are displayed accurately, with occupied slots excluded.
- A booking session completes successfully for a chosen available slot.
- The confirmed session is added to both the student's and mentor's calendars with a valid meeting link.
- The booked slot is removed from future availability listings.

9. Acceptance / Pass-Fail Criteria

PASS: The test case passes if mentor recommendations correctly reflect domain matching, available slots are displayed accurately, the booking is completed successfully, and the confirmed session is added to both calendars with a valid meeting link.

FAIL: The test case fails if non-matching mentors are recommended, unavailable or incorrect slots are displayed, the booking is not created, or the session is missing from either calendar or does not contain a valid meeting link.

10. Test Status

Execution Status	Not Executed
Reason	This report specifies the test case and expected results. No system implementation is being tested as part of this report.
Final Result	Pending / Not Applicable until implementation is available

11. Use Case Diagram

